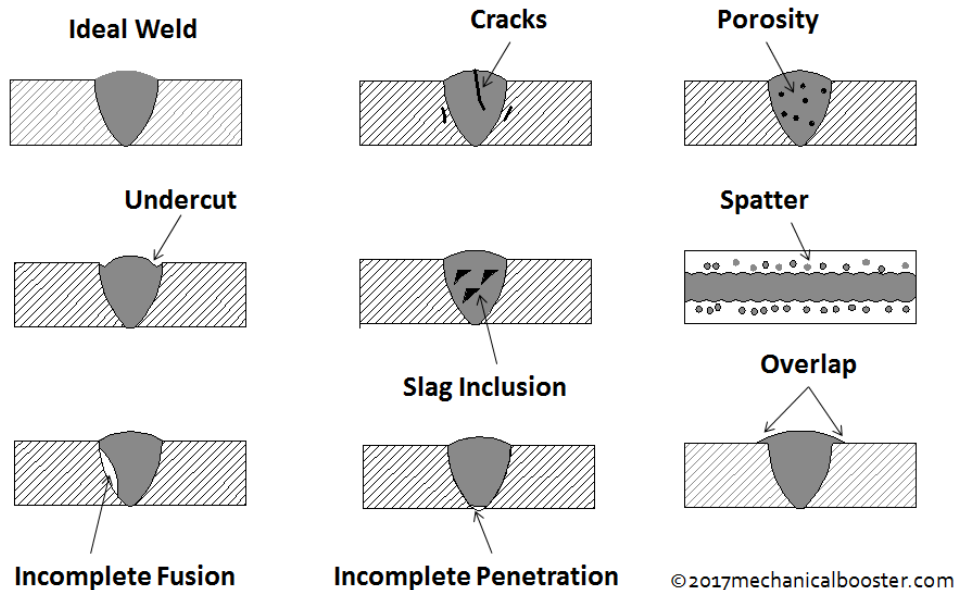


A. Welding Defects:

Welding Defects can be defined as the irregularities formed in the given weld metal due to wrong welding process or incorrect welding patterns, etc. The defect may differ from the desired weld bead shape, size, and intended quality. Welding defects may occur either outside or inside the weld metal. Some of the defects may be allowed if the defects are under permissible limits but other defects such as cracks are never accepted.

Different Welding Defects



Welding defects can be classified into two types as external and internal defects:

External Welding Defects:

1. Weld Crack
2. Undercut
3. Spatter
4. Porosity
5. Overlap
6. Crater

Internal Welding Defects:

1. Slag Inclusion
2. Incomplete Fusion
3. Necklace cracking
4. Incompletely filled groove or Incomplete penetration

External Welding Defects

The various types of external defects with their causes and remedies are listed below:

1. Weld Crack

This is the most unwanted defect of all the other welding defects. Welding cracks can be present at the surface, inside of the weld material or at the heat affected zones.

Crack can also appear at different temperatures:

Hot Crack – It is more prominent during crystallization of weld joints where the temperature can rise more than 10,000-degree Celsius.

Cold Crack – This type of crack occurs at the end of the welding process where the temperature is quite low. Sometimes cold crack is visible several hours after welding or even after few days.

Causes Of Weld Crack:

1. Poor ductility of the given base metal.
2. The presence of residual stress can cause a crack on the weld metal.
3. The rigidity of the joint which makes it difficult to expand or contract the metals.
4. If there is high content on sulfur and carbon then also the cracks may appear.
5. Using hydrogen as a shielding gas while welding ferrous materials.

Remedies for Weld crack:

1. Using appropriate materials may decrease the chances of crack.
2. Preheating the weld and reducing the cooling speed joint helps in reducing crack.
3. Reduce the gap between the weld joints by using reasonable weld joints.
4. While welding releases the clamping force slowly which increases fill to capacity of welding material.

2. Undercut

When the base of metal melts away from the weld zone, then a groove is formed in the shape of a notch, then this type of defect is known as Undercut. It reduces the fatigue strength of the joint.

Causes of Undercut:

1. If the arc voltage is very high then this defect may occur.
2. If we use the wrong electrode or if the angle of the electrode is wrong, then also the defect may form.
3. Using a large electrode is also not advisable.
4. High electrode speed is also one of the reasons for this defect.

Remedies for Undercut:

1. Reduce the arc length or lower the arc voltage.
2. Keep the electrode angle from 30 to 45 degree with the standing leg.
3. The diameter of the electrode should be small.
4. Reduce the travel speed of the electrode.

3. Spatter

When some metal drops are expelled from the weld and remain stuck to the surface, then this defect is known as Spatter.

Causes Of Spatter:

1. High Welding current can cause this defect.
2. The longer the arc the more chances of getting this defect.
3. Incorrect polarity.
4. Improper gas shielded may also cause this defect.

Remedies for Spatter:

1. Reducing the arc length and welding current
2. Using the right polarity and according to the conditions of the welding.
3. Increasing the plate angle and using proper gas shielding.

4. Porosity

Porosity in the condition in which the gas or small bubbles gets trapped in the welded zone.

Causes of Porosity:

1. It occurs when the electrode is not coated properly.
2. Using a longer arc may also increase its chances.
3. Increased welding currents.
4. Rust or oil on the welding surface.

Remedies for porosity:

1. Proper selection of the electrode.
2. Decreasing the welding current.
3. Using smaller arc and slowing the process to allow the gases to escape.
4. Remove rust or oil from the surface and use a proper technique.

5. Overlap

When the weld face extends beyond the weld toe, then this defect occurs. In this condition the weld metal rolls and forms an angle less than 90 degrees.

Causes of Overlap:

1. Improper welding technique.
2. By using large electrodes this defect may occur.
3. High welding current

Remedies for Overlap:

1. Using a proper technique for welding.
2. Use small electrode.
3. Less welding current.

6. Crater

It occurs when the crater is not filled before the arc is broken, which causes the outer edges to cool faster than the crater. This causes a stress and then crack is formed.

Causes of the crater:

1. Incorrect torch angle.
2. Use of large electrode:
3. Improper welding technique

Remedies for crater:

1. Using a proper torch angle may reduce the stress on the metal
2. Using a small electrode may also decrease the crater.
3. Use a proper technique.

Internal Welding Defects

The various types of internal welding defects with their causes and remedies are listed below:

1. Slag Inclusion

If there is any slag in the weld, then it affects the toughness and metal weldability of the given material. This decreases the structural performance of the weld material. Slag is formed on the surface of the weld or between the welding turns.

Causes Of Slag:

1. Slag is formed if the welding current density is very small, as it does not provide the required amount of heat for melting the metal surface.
2. If the welding speed is too fast then also slag may occur.
3. If the edge of the weld surface is not cleaned properly then also slag may form.
4. Improper welding angle and travel rate of welding rod.

Remedies for Slag Inclusion:

1. Increase the current density
2. Adjust the welding speed so that the slag and weld pool do not mix with each other.
3. Clean the weld edges and remove the slags of previous weld layers
4. Have a proper electrode angle and travel rate.

2. Incomplete Fusion

Incomplete fusion occurs when the welder does not accurately weld the material and the metal pre solidifies which leads to a gap which is not filled with the molten metal.

Causes of Incomplete fusion:

1. It occurs because of the low heat input.
2. When the weld pool is very large and runs ahead of the arc.
3. When the angle of the joint is too low.
4. Incorrect electrode and torch angle may also lead to incomplete fusion.
5. Unproper bead position.

Remedies for Incomplete Fusion:

1. Increasing the welding current and decreasing the travel speed helps in removing the chances of incomplete fusion.
2. Reducing the deposition rate.
3. Increasing the joint angle.

4. Try to position the electrode and torch angle properly so that the edges of the plate melt away.
5. Positioning the bead properly so that the sharp edges with other beads can be avoided.

3. Necklace Cracking

It occurs in the use of electron beam welding where the weld does not penetrate properly. Therefore, the molten metal does not flow into the cavity and results in a cracking known as “Necklace Cracking”.

Causes of Necklace Cracking:

1. Improper welding technique.
2. It occurs in materials such as nickel base alloys, stainless steel, carbon steels and Tin alloys.
3. Using high speed of electron beam welding

Remedies for Necklace Cracking:

1. Using a proper welding technique reduce the chances of necklace cracking.
2. Using proper materials for welding.
3. Using a constant speed during the welding process.
3. Improper welding technique

4. Incompletely Filled Groove or Incomplete Penetration

These defects occur only in the butt welds where the groove of the metal is not filled completely. It is also called as incomplete penetration defect.

Causes of an Incomplete filled groove are:

1. Less deposition of the weld metal
2. Use of improper size of the electrode
3. Improper welding technique

Remedies for Incomplete filled groove are:

1. More deposition of the weld metal.
2. Use a proper size of the electrode.
3. By using a proper welding technique.

B. Laser welding

Laser beam welding (LBW) is a welding technique used to join pieces of metal or thermoplastics through the use of a laser. The beam provides a concentrated heat source, allowing for narrow, deep welds and high welding rates.

A **continuous or pulsed** laser beam may be used depending upon the application.

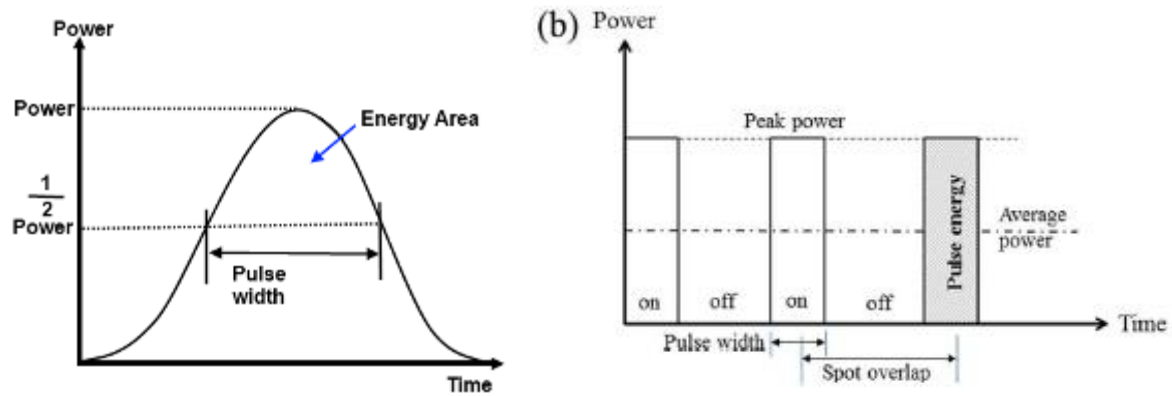


Figure: 2 Pulse characteristics (a) Sine curve; (b) Square pulse

The **process parameters** are Laser power, welding speed, spot diameter, spot overlap, focal length, power density for continuous laser and peak power, pulse width, average power, welding speed, spot diameter, spot overlap, focal length, power density.

Laser beam welding has high power density (on the order of 1 MW/cm^2) resulting in small **heat-affected zones** and high heating and cooling rates. The spot size of the laser can vary between 0.2 mm and 13 mm, though only smaller sizes are used for welding. The depth of penetration is proportional to the amount of power supplied, but is also dependent on the location of the **focal point**: penetration is maximized when the focal point is slightly below the surface of the workpiece

The efficiency of laser beam process is low as most of the beam gets reflected and transmitted and only a small amount is absorbed. The optical properties like reflectivity, absorptivity.

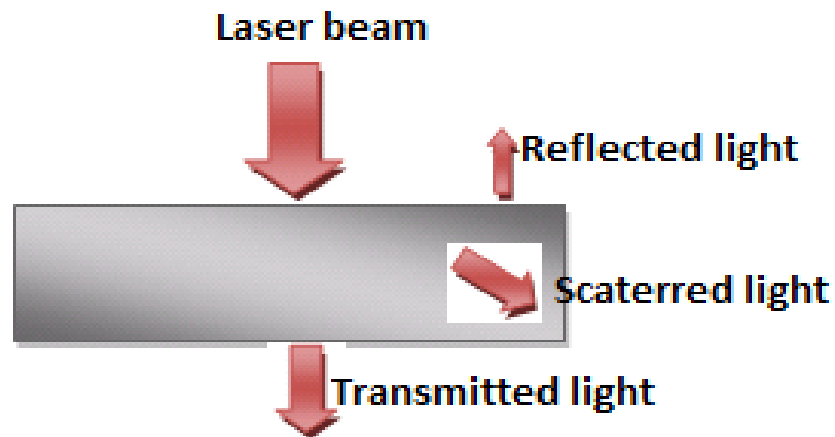


Figure 3: Optical properties of the base material

There are **three basic types of lasers** viz., the solid- state laser (especially **ruby lasers** and **Nd:YAG lasers**), the gas laser (**helium, nitrogen, and carbon dioxide**); and the semi-conductor laser.

Advantages:

- ✓ The weld is formed as the intense laser light rapidly heats the material-typically calculated in milli-seconds.
- ✓ Laser welding is a non-contact process ,very little distortion of final joint geometry, narrow heat effected zone
- ✓ Its high precision, as no additional adhesive materials are required.
- ✓ Precisely controlled intense heat source.

Disadvantages:

- ✓ Initial cost is high. The equipment used in LBW has high cost.
- ✓ High maintenance cost.
- ✓ Due to rapid rate of cooling, cracks may be produced in some metals.
- ✓ High skilled labour is required to operate LBW.
- ✓ The welding thickness is limited to 19 mm.
- ✓ The energy conversion efficiency in LBW is very low. It is usually below 10 %.

Principle and Mechanism of Laser Welding:

The function of a laser is to amplify light. Ordinary light cannot be used as a laser light because radiant energy from an ordinary light source is incoherent and distributed over a wide spectral range, and monochromatic single- colour sources do not exist. Due to variant wavelengths of different colours constituting ordinary light it is not possible to collimate it to a sharp focus without sacrificing intensity.

Thus laser works on the principle that when electrons of an atom gets excited by absorbing some energy. This excited state is short-lived and the atom immediately drops back to an intermediate level or metastable state. In this dropping back the atom loses its heat energy but retains its photon energy. Soon after the atom falls spontaneously and randomly back to the ground state releasing the photon energy, or quantum energy, in the form of light as depicted in Fig. 3. This automatic dropping back to original energy level, without being stimulated to do so, is referred to as spontaneous emission.-

Laser Pumping Mechanism at Atomic Level

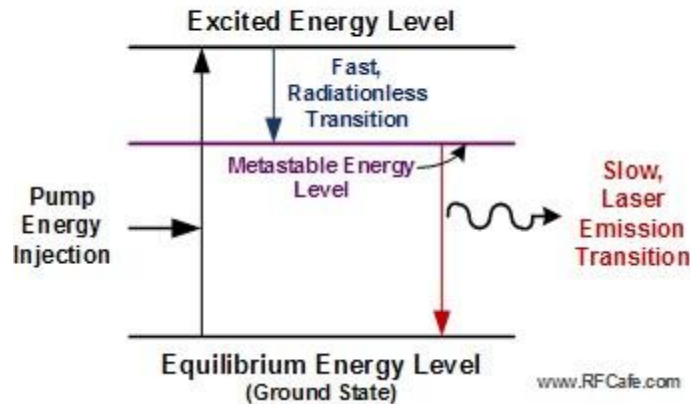


Figure 4: Mechanism of laser emission

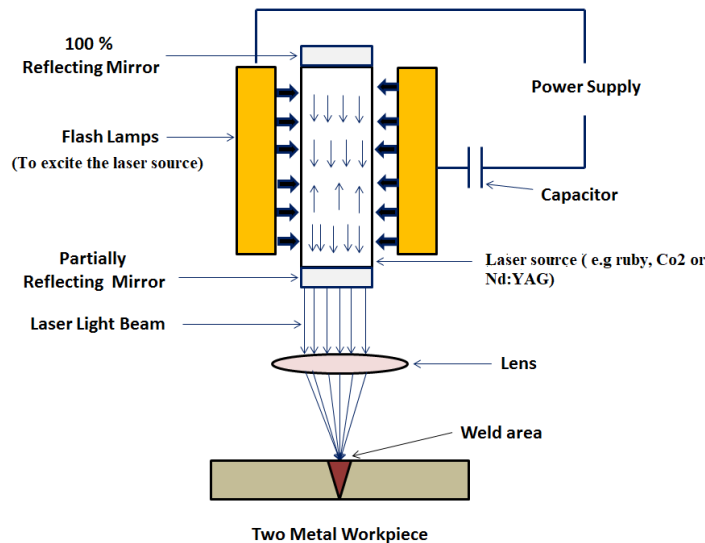


Figure 5: Experimental setup of laser welding.

Main Parts of experimental paramters

The main parts or equipment of laser beam welding are:

1. **Laser Machine:** It is a machine that is used to produce laser for welding. The main components of laser machine are shown below.
2. **Power Source:** A high voltage power source is applied across the laser machine to produce laser beam.
3. **CAM:** It is a computer aided manufacturing in which the laser machine is integrated with the computers to perform welding process. All the controlling action during the welding process by laser is done by CAM. It speeds up the welding process to a greater extent.
4. **CAD:** It is called as Computer aided Design. It is used to design the job for welding. Here computers are used to design the workpiece and how the welding is performed on it.

5. **Shielding Gas:** A shielding gas may be used during the welding process in order to prevent the w/p from oxidation.

C. Electron beam welding:

Electron-beam welding (EBW) is a fusion welding process in which a beam of high-velocity electrons is applied to two materials to be joined. The workpieces melt and flow together as the kinetic energy of the electrons is transformed into heat upon impact. EBW is often performed under vacuum conditions to prevent dissipation of the electron beam.

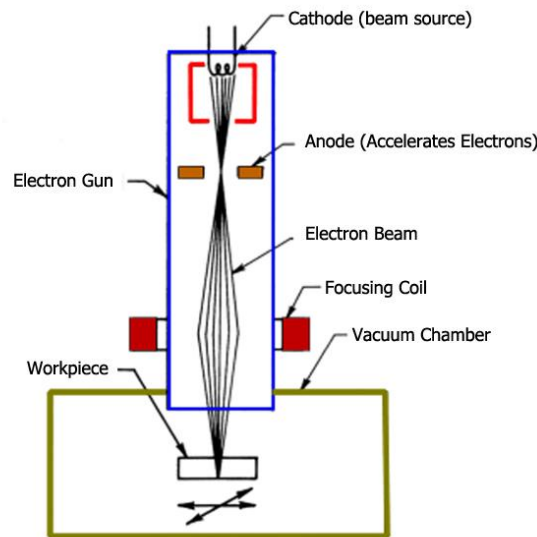


Figure 6: Experimental setup of electron beam welding.

Principle of electron beam Welding

Electrons are elementary particles possessing a mass $m = 9.1 \cdot 10^{-31}$ kg and a negative electrical charge $e = 1.6 \cdot 10^{-19}$ C. They exist either bound to an atomic nucleus, as conduction electrons in the atomic lattice of metals, or as free electrons in vacuum.

Free electrons in vacuum can be accelerated, with their paths controlled by electric and magnetic fields. In this way narrow beams of electrons carrying high kinetic energy can be formed, which upon collision with atoms in solids transform their kinetic energy into heat. Electron-beam welding provides excellent welding conditions because it involves:

- Strong electric fields, which can accelerate electrons to a very high speed. Thus, the electron beam can carry high power, equal to the product of beam current and accelerating voltage. By increasing the beam current and the accelerating voltage, the beam power can be increased to practically any desired value.

- Using magnetic lenses, by which the beam can be shaped into a narrow cone and focused to a very small diameter. This allows for a very high surface power density on the surface to be welded. Values of power density in the crossover (focus) of the beam can be as high as $10^4 - 10^6 \text{ W/mm}^2$.
- Shallow penetration depths in the order of hundredths of a millimeter. This allows for a very high volumetric power density, which can reach values of the order $10^5 - 10^7 \text{ W/mm}^3$. Consequently, the temperature in this volume increases extremely rapidly, $10^8 - 10^{10} \text{ K/s}$.

Process Parameters:

The size and shape of the zone influenced by the beam depends on:

- (1) Beam power – The power of the beam [W] is the product of the accelerating voltage [kV] and beam current [mA], parameters easily measurable and precisely controllable. The power is controlled by the beam current at constant accelerating voltage, usually the highest accessible.
- (2) Power density (focusing of the beam) – The power density at the spot of incidence of the beam with the workpiece depends on factors like the size of the electron source on the cathode, the optical quality of the accelerating electric lens and the focusing magnetic lens, alignment of the beam, the value of the accelerating voltage, and the focal length. All these factors (except the focal length) depend on the design of the machine.
- (3) Welding speed – The construction of the welding equipment should enable adjustment of the relative speed of motion of the workpiece with respect to the beam in wide enough limits, e.g., between 2 and 50 mm/s.
- (4) Material properties, and in some cases also on
- (5) Geometry (shape and dimensions) of the joint.

The final effect of the beam depends on the particular combination of these parameters.

- Action of the beam at low power density or over a very short time results in melting only a thin surface layer.
- A defocused beam does not penetrate, and the material at low welding speeds is heated only by conduction of the heat from the surface, producing a hemispherical melted zone.
- At high power density and low speed, a deeper and slightly conical melted zone is produced.
- In the case of very high power density, the beam (well focused) penetrates deeper, in proportion to its total power.

Advantages of Electron Beam Welding

1) A clean process as it is carried out in a vacuum chamber:

EBW is free from potential gas contamination as the welding is done in a vacuum chamber, resulting in a high-quality weld with no oxidation.

2) Good for delicate assemblies:

Since EBW uses localised heating (small Heat Affected Zone, or HAZ) and has minimal heat input, the surrounding material stays cool. This allows for welding of delicate products and close to heat sensitive areas such as populated electronics enclosures. EBW has a HAZ similar to laser beam welding, which is less than 5% of a similar TIG weld.

3) High quality and integrity from machine-controlled automation:

EBW has high strength with minimal distortion or damage. Also, very little post-weld machining is needed due to its accuracy, making it suitable for welding precision parts. Since the machine is programmable, it is possible to have custom weld-profiles. Our quality is assured through AS9100 Rev D, AS9104 and NADCAP certification as well as bespoke OEM approvals.

4) Repeatable accuracy and quality control:

Precise parameters can be machine-programmed to reproduce welds in large quantities whilst maintaining accuracy.

5) Joining dissimilar materials:

One of EBW's greatest advantages is its ability to weld materials with different melting points, such as copper to stainless steel. From a cost and performance perspective EB welding also has the ability to weld high-cost nickel alloys to relatively low-cost stainless steel.

For example, the corrosive performance of a nickel alloy which is only required in a particular area within a product's design, can be EB welded to a low-cost stainless steel which is not subjected to the same corrosive environment which the nickel alloy is. Heat-treated components can be EB welded as can highly reflective metals such as platinum. Refractory metals such as titanium can also be EB welded.

6) No filler material used:

Typically, no filler materials are used in the EB welding process. It is the parent material which is melted to create the join and thus the assembly.

7) Range of penetrations:

A variety of weld penetrations are possible, from 0,005" (0,127mm) to 2" (50mm). It also has a high depth to width ratio (10:1 to 40:1 in many cases).

8) High welding speeds:

Although EBW has a longer cycle time than laser beam welding, it achieves high welding speeds of up to 30mm per second.

9) Cost Savings:

EBW is a cost-effective method of welding, only requiring a single pass of the electron beam, unlike TIG, MIG and arc welding which tend to endorse multiple passes of the heat source. The EBW process can also deliver weight savings if the products design allows for this, which can then lead on to further benefits such as a reduction in lead time and inventory costs.

Disadvantages of Electron Beam Welding:

For a balanced overview let's look at a few reasons why EBW may not be suitable...

1) Setup costs:

The initial outlay for equipment is expensive. With these costs in mind, a company will need a large work-flow in order to justify the investment. It is usually more cost effective to outsource your EBW requirements unless you have a large volume throughput over a long period of time.

2) Equipment:

Due to the complex technical equipment, expert operators are essential. If you are starting with a clean piece of paper, you will need to invest in professional training and employ experienced qualified welding engineers.

3) Size limitations:

The size of the components to be welded is limited by the size of the vacuum chamber at your disposal.

4) X-Rays & Radiation:

Safety standards are exceptionally strict as x-rays and radiation are both present during the welding process.